

Firefly Reading Circle

Building Advanced Nuclear Systems in India

Purpose

A monthly technical reading circle for engineers, researchers, students, and builders interested in advanced nuclear energy, reactor design, systems engineering, and deep-tech innovation.

Vision

Create a community that evolves from a reading group into a study group and open-source simulation community focused on advanced nuclear systems.

Format (Monthly, 2 Hours)

- 30 min Technical Overview
- 60 min Guided Discussion
- 30 min Hands-on Workshop

Each session produces shared notes, technical summaries, simulation exercises, and research questions.

2026 Founding Series

Month 1 – Nuclear Engineering Fundamentals

Reading: Lamarsh + OpenMC tutorials

Workshop: Install Python, Jupyter, OpenMC and run first simulation.

Month 2 – Reactor Architectures

Reading: Stacey + Generation IV Roadmap

Workshop: Reactor comparison matrix and technology landscape report.

Month 3 – Fast Reactors & Fuel Breeding

Reading: Fast Breeder Reactors + ORNL MSR reports

Workshop: Compare thermal vs fast reactor simulations in OpenMC.

Month 4 – Molten Salt Systems Engineering

Reading: Molten Salt Reactors and Thorium Energy + ORNL MSRE documentation

Workshop: Develop a complete MSR system architecture.

Tools We Will Learn

OpenMC, OpenFOAM, Python, Jupyter, FreeCAD, GitHub, systems modeling.

Who Should Join?

Mechanical, Chemical, Materials, Electrical and Software Engineers, Physicists, Researchers, Students, and Deep-Tech Builders.